

Dealer Attention Engine: Data to Decision

4 October 2026

The problem

Most dealers are judged on one number: sales against target. It reports trouble late and often blames the wrong dealer.

Networks are too big to visit. India's largest car maker had 3,863 sales outlets in 2,522 cities in FY24, over 2,000 of them in non-urban markets ([Motilal Oswal, from the FY24 annual report](#)).

What is assumed	Why it fails
Low sales = bad dealer	The market may be small. The dealer may be the best operator there.
High sales = good dealer	Targets can be hit by pushing stock, deep discounts, late payments or losing customers.
Act when sales drop	Sales move last. Payments, stock age and complaints move first.

The reframe: sales says what already went wrong. This engine says who is next, why, and the one action to take.

Size of the prize

About **Rs 23,300 crore** of vehicle inventory value may sit above FADA's 21-day benchmark across the industry (August 2026).

Step	Working	Result	Source
Stock held	FADA estimate, Aug 2026	38 to 40 days	FADA
Norm	FADA recommendation	21 days	FADA
Excess	38 to 40 minus 21	17 to 19 days (mid 18)	Derived
Retail per day	4,02,398 PV retail ÷ 31 days	12,981 cars	FADA
Excess cars	12,981 × 18	2.34 lakh (2.21 to 2.47)	Derived
Value per car	Rs 77,800 cr ÷ 7.8 lakh cars, Aug 2024	Rs 9.97 lakh	FADA
Inventory value above norm	2.34 lakh × Rs 9.97 lakh	Rs 23,300 cr (22,000 to 24,600)	Derived

Read with care

- **Use retail, not wholesale.** August PV wholesale was 4.39 lakh; inventory days are counted on retail ([ICICIdirect](#)).
- **Value per car is from 2024.** Car prices fell with the September 2025 GST cut ([Deccan Herald](#)), so Rs 9.97 lakh is likely high today.
- **August is a seasonal high.** Festive stocking had begun. For the steady-state excess, average the last 12 monthly FADA readings.
- **Stock is uneven by maker.** In August, Maruti held about 16 days and Hyundai 4 to 5 weeks ([ICICIdirect](#)). Never split the industry figure by market share.
- **Inventory value is not the loss.** The yearly cost is the interest on it, at the client's floor plan rate. Most of it is dealer and bank money.

The client's prize uses only the client's own numbers:

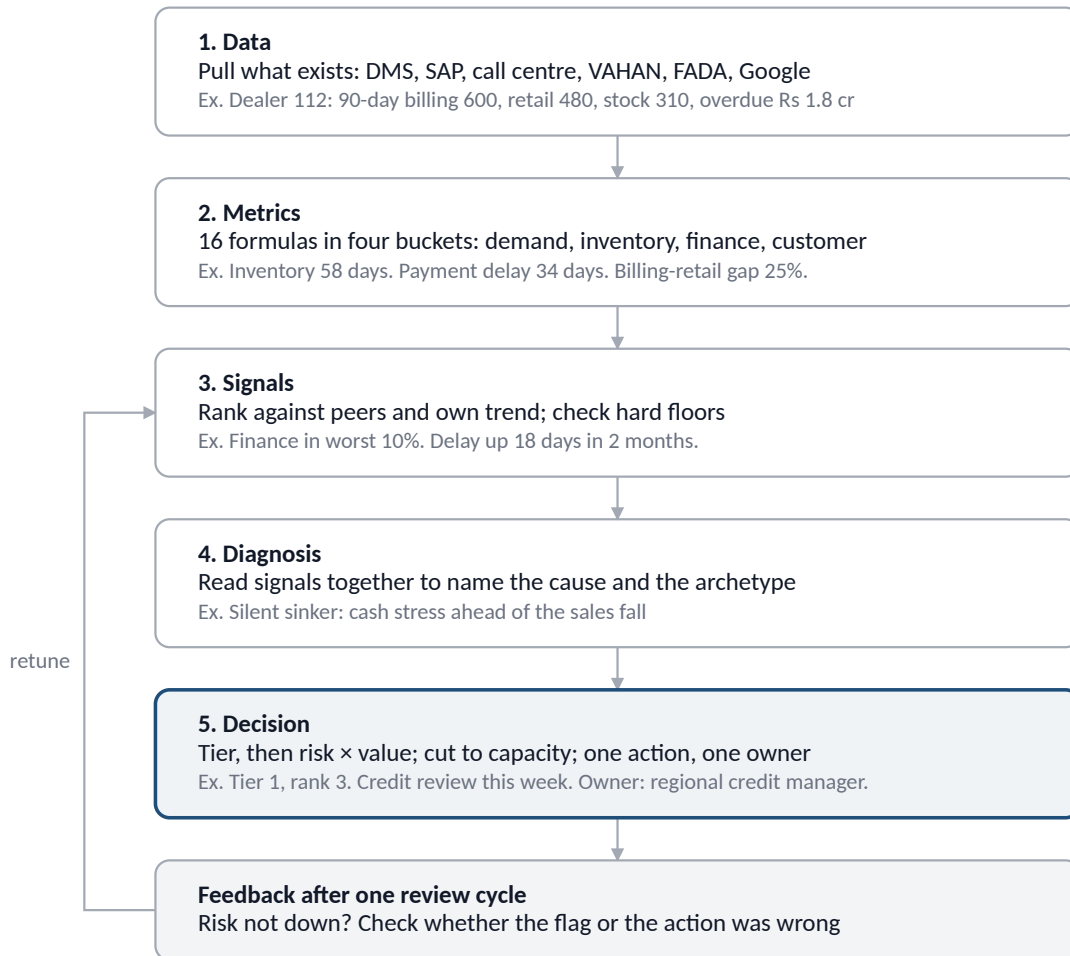
$$\text{Yearly prize} = (\text{Days held} - \text{Stocking norm}) \times \text{Retail per day} \times \text{Value per car} \times \text{Floor plan rate}$$

The engine claims only stock above the norm at the dealers it flags. Peer ranking cannot see an industry-wide excess, so inventory also gets a hard floor (Step 3).

The chain

Each step narrows the data until one dealer, one cause, one action and one owner remain.

Five steps turn data the OEM already has into one ranked action list



Data to decision: 5 steps plus feedback

Step 1. Data

Rule: every field names the system it already lives in. No source, no field.

Tier 1: already inside the OEM

Data	System
Retail sales, model, price, discount	DMS
Enquiries, test drives, bookings, lost reasons	DMS or CRM
Wholesale billing to dealer	OEM ERP (SAP)
Stock by model, billing date per VIN	DMS + OEM dispatch
Receivables, due date, paid date, overdue	OEM ERP finance

Data	System
Credit limit, floor plan use	OEM finance + financing bank
Service jobs, labour and parts revenue, first-time fix	DMS service
Complaints, severity, days to close, escalation	Call centre or CRM
CSAT or NPS	OEM survey system
Sales headcount and exits	Dealer portal, field app
Dealer master: district, outlet type, age	Network management

Tier 2: public

Source	What it gives
VAHAN (MoRTH)	Monthly registrations by RTO and maker. Market size and brand share by district. The only source for the denominator.
FADA	Industry benchmarks: inventory days, dealer liquidity. August 2026 data covered 1,467 of 1,469 RTOs (Auto Punditz).
Google Maps reviews	Rating, count, text, response rate per outlet. Not dealer-reported.

Tier 3: not collected today. Mystery shopping, dealer P&L, competitor street prices. Out of version 1. Phase 2 collects them only for dealers version 1 flags, which keeps cost low.

When sources disagree

Level	Trust order
District market	VAHAN > OEM ERP > third party > dealer-declared
Single dealer	DMS > OEM ERP > dealer-declared. VAHAN is never used here.

Social media: excluded from scoring. An outlet sells about 43 cars a month (1,65,200 Maruti retail in August 2026 ÷ 3,863 outlets; [Autocar Pro](#)), and fewer as the network grows. Too few posts name a single dealer. Used only as an escalation tripwire and for brand-level mood.

Data limits

Issue	Effect	Handling
VAHAN coverage	The public Vahan 4 dashboard may still miss Telangana (Flash Finance)	Check each pull. Show state coverage on every output.
VAHAN geography	A car registers at the buyer's RTO, not the dealer's	Market size only, as a directional benchmark: RTO-based, not true dealer catchment. Never dealer attribution.
VAHAN timing	Registration lags the sale; figures get revised	Rolling windows. Timestamp every pull.
Google volume	Rural outlets get few reviews	Google is a cross-check only, used above a minimum review count. The customer bucket still scores from internal data.
Google bias	Reviews asked for at delivery skew high	Compare within cohort, never absolute.
Dealer-declared fields	Can be gamed	Hints only, never the deciding input.

Step 2. Metrics

Raw fields become 16 metrics in four buckets. Target achievement stays as a context KPI with a low weight, so it cannot dominate the early-warning score.

Bucket	Metric	Formula
Demand	Target achievement (context, low weight)	Retail ÷ target
	Growth	Retail, last 3 months ÷ same 3 months last year
	District share	Dealer retail ÷ district registrations in segment
	Potential capture (equal-share proxy)	Dealer retail ÷ (district registrations × brand's state share ÷ brand outlets in district)
	Funnel conversion	Retail ÷ enquiries
	Discount intensity	Discount ÷ ex-showroom price (%)
Inventory	Inventory days	Closing stock ÷ average daily retail, last 90 days
	Ageing	Units over 60 days ÷ total stock
	Mix mismatch	Gap between stock model mix and district model mix (brand's own DMS)
	Billing-retail gap	(Wholesale – retail) ÷ retail, last 3 months. Not push on its own: festive stocking, launches and transit also widen it.
Finance	Weighted payment delay	Sum of (overdue amount × days past due) ÷ total overdue
	Overdue ratio	Overdue ÷ total receivable
	Credit stress	Floor plan use (%) and bounce count, ranked separately, then averaged

Bucket	Metric	Formula
Customer	Complaint rate	Sales complaints per 1,000 retails and service complaints per 1,000 service jobs, 12 months, ranked separately
	Service retention index	Cars sold 12 to 36 months ago that came back for service in the last 12 months ÷ cars sold 12 to 36 months ago
	Resolution speed	Median days to close and % escalated, ranked separately

Why these choices

Choice	Reason
Growth on 3 months against last year	Removes seasons, and festivals that shift month to month. FADA linked August 2026's dip partly to that shift (FADA).
Service retention on cars 12 to 36 months old	Cars under 12 months are still in free service. The same cars sit in top and bottom, so it cannot pass 100%. Read only against peers, as mix and geography also move it.
Ratios, not units or rupees	Big dealers and SUV-heavy dealers are not punished for size or mix.
Potential split equally by brand outlets	Version 1 assumes equal opportunity across same-brand outlets in a district; version 2 would use catchment or lead data. State share is fairer than national share.
Mixed units ranked separately	A % and a count cannot be added.

Diagnosis-only inputs (not scored): staff exits, enquiry volume, district registration trend, CSAT or NPS, Google rating.

Watch for gaming: enquiries are dealer-logged. A conversion jump with no sales change usually means logging stopped. Cross-check against OEM-owned digital leads.

Step 3. Signals

A metric becomes a signal when it is bad against peers, getting worse, or past a hard limit.

Trigger	Rule	Why
Level	Percentile within the peer cohort	A small-town dealer is judged against small towns
Trend	Change over a rolling window, also as a percentile	Fires while the MIS report still looks fine
Hard floor	Past the credit period, sanctioned limit or stocking norm in the dealer agreement	Catches a problem that hits all peers at once, which percentiles miss

Peer cohort = district market size, outlet type, dealer age, number of same-brand outlets in the district.

Each signal carries its direction and how many months in a row it has fired. The client sets how many months turn a wobble into a pattern.

Every number is one of four kinds

Kind	Meaning	Example
Published	External figure, cited	FADA 21-day norm
Derived	Arithmetic on published figures, working shown	Rs 9.97 lakh per car
Contractual	From the client's dealer agreement or financing terms	Credit period, stocking norm
Calibrated	The client sets the value; we give the mechanism	Flag cutoff, weights, windows

Where each threshold comes from

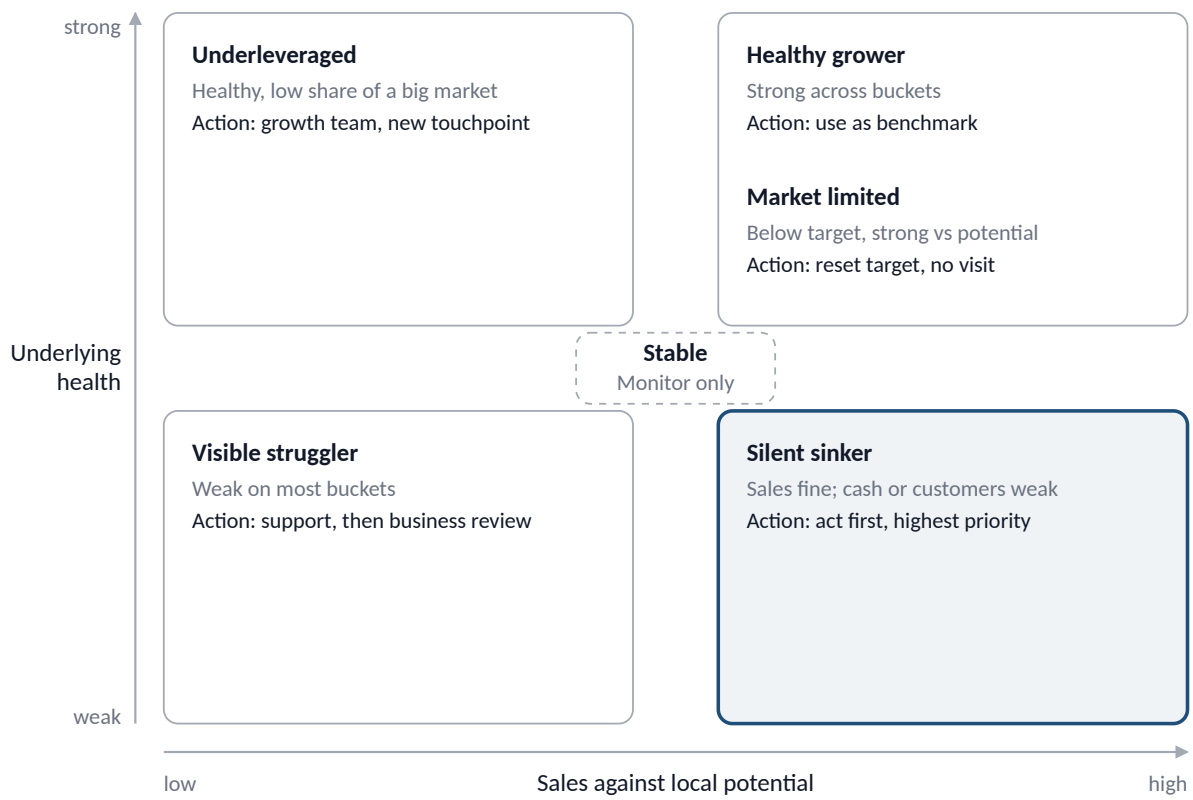
Metric	Threshold	Kind
Inventory days	FADA's 21 days sizes the industry excess. The client's stocking norm is the hard floor. Flags on cohort percentile.	Published, contractual, calibrated
Ageing	60 and 90 days are placeholders until the dealer agreement gives the funding terms	Contractual (pending)
Payment delay, overdue	Days past the credit period in the dealer agreement	Contractual
Floor plan use	Against the sanctioned limit	Contractual
Share, growth, conversion, discount	No Indian benchmark exists. Cohort percentile.	Calibrated
Complaints, resolution, service retention index	No Indian benchmark exists. Cohort percentile, Google as cross-check.	Calibrated

No US benchmarks. NADA's guide recommends 45 to 60 days of new-car stock ([Dealer Marketing](#)); FADA recommends 21. A US-tuned model would call a 45-day Indian dealer healthy, at more than double FADA's norm.

Step 4. Diagnosis

One signal is noise. Signals that fire together name a cause, and the cause names the action. Every dealer lands in one of six archetypes each month. Each diagnosis is a hypothesis the manager confirms on the ground.

The silent sinker hides where sales look healthy



Six archetypes: sales against potential vs finance, stock and customer health

When a dealer fits two, the more urgent wins: silent sinker, visible struggler, underleveraged, market limited, stable, healthy grower.

Signals firing together	Cause	Archetype	Action
Billing up, retail flat, stock days and ageing up	Stock pushed, not sold	Silent sinker	Pause billing, move or buy back aged stock, fix model mix
Delay up, overdue up, floor plan near limit, retail normal	Cash stress ahead of a sales fall	Silent sinker	Credit review, revised limit, escalate early
Discount up, volume flat	Buying volume, losing margin	Silent sinker	Pricing discipline, scheme audit
Complaints up, slower resolution, retention down	Brand damage, future sales loss	Silent sinker	Service audit, workshop capacity, training
Enquiries steady, conversion down, staff exits up	Execution and staffing gap	Visible struggler	Hiring support, sales training, incentive fix
Weak on most buckets for months, no response to past help	Structurally unviable	Visible struggler	Business review: turnaround, restructure or exit decision

Signals firing together	Cause	Archetype	Action
Enquiries down, district registrations down too	Market shrinking	Market limited	Reset target. No visit.
Retail healthy, share low, potential high	Room to grow	Underleveraged	Add touchpoint, growth team

The silent sinker justifies the build. Today's reports cannot see it, because sales look fine.

Step 5. Decision

The list is ordered in two passes: tier first, then risk × value. It is cut at what one manager can act on in a month.

Pass 1: tier. Any bucket in the worst 10% of its cohort, or any hard floor breached, goes to Tier 1: relative underperformance or absolute breach. The 10% cut is a placeholder the client sets.

Why the tier comes first: an average hides a single deep problem.

Dealer	Finance	Inventory	Demand	Customer	Risk score	Rank by score alone	Rank with tier
A, silent sinker	100	30	30	30	54.5	2nd	1st
B, mediocre everywhere	65	65	65	65	65	1st	2nd

Pass 2: risk × value.

$$\text{Risk} = 0.35 \times \text{Finance} + 0.25 \times \text{Inventory} + 0.25 \times \text{Demand} + 0.15 \times \text{Customer}$$

- Bucket score = average risk percentile of its metrics (0 best, 100 worst). Target achievement carries a low weight inside Demand.
- Weights are illustrative. Finance weighs most as the survival bucket that moves first. Customer weighs least for immediate survival, but it is a leading signal of future demand and brand damage, and its worst 10% still triggers Tier 1. The client's lost-dealer record will calibrate all four.

$$\text{Value at stake} = \text{Annual retail} \times \text{Share of buyers lost} \times \text{Contribution per car} + \text{Service revenue at risk}$$

- **Share of buyers lost:** many buyers of a failed dealer move to the next brand dealer. Shown as 25%, 50% and 75% scenarios until the OEM's past dealer exits calibrate it.
- **Contribution per car (client input):** illustrative placeholder of Rs 60,000, not a validated floor. It is Maruti FY26 net profit of Rs 14,445 cr (*The Machine Maker*) ÷ 24.23 lakh cars sold (*Maruti*), used only for order of magnitude. Net profit per car is not contribution per car, so the client supplies the real figure (question 8).

One output row (Dealer 112 is a labelled representative extract)

Field	Value
Dealer	Dealer 112, Nashik
Archetype	Silent sinker
Tier and rank	Tier 1 (finance in worst 10%), rank 3 this month
Bucket scores	Finance 98, Inventory 89, Demand 74, Customer 61
Risk score	$0.35 \times 98 + 0.25 \times 89 + 0.25 \times 74 + 0.15 \times 61 = \mathbf{84}$
Value at stake (illustrative)	At 50% lost: 1,920 cars a year $\times$ 50% $\times$ Rs 60,000 = Rs 5.8 cr a year . At 25% to 75% lost: Rs 2.9 to 8.6 cr. Plus service revenue.
Retail	160 a month, steady for 3 months
Top signals	Payment delay 34 days, up 18 in 2 months. Floor plan at 94%. Stock 58 days, 31% over 60 days old. Billing 25% above retail.
Diagnosis	Cash stress ahead of a sales fall
Action	Credit review and revised limit this week. Hold billing. Move aged stock.
Owner	Regional credit manager
Data coverage	All four buckets scored

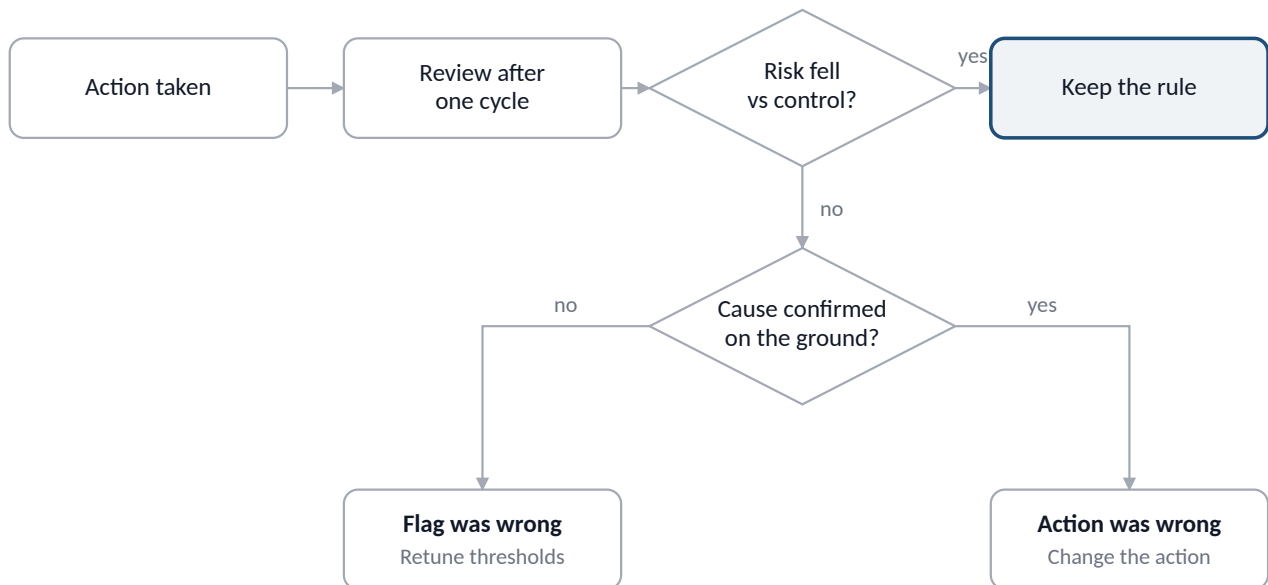
Check the numbers: 90-day retail 480, so 5.33 cars a day. Stock $310 \div 5.33 = 58$ days. 96 of 310 units over 60 days = 31%. Billed 600 against 480 retailed = 25% billing-retail gap. $160 \times 12 = 1,920$ cars a year.

Feedback loop

Each flagged dealer is tracked for one review cycle after the action. The client sets the cycle length (calibration question 6).

When risk does not fall, find out whether the flag or the action failed

Control = flagged dealers just below the cutoff that were not visited



Feedback loop: 2 checks, 3 outcomes

Why a control group: dealers flagged at the extremes drift back toward average on their own, so every action would look like it worked. Dealers just below the capacity cutoff are similar but were not visited, which gives a free comparison.

Where ML fits

Version 1 runs on percentiles and rules, so every flag can be explained to a manager and defended to a dealer. ML enters only where rules cannot do the job.

Technique	Used on	Why it earns its place
NLP (aspect sentiment, topics)	Google review and complaint text	Splits one star rating into sales, service, paperwork and finance. Rules cannot read text.
Supervised model	Bucket weights	Learns weights from real lost dealers. Caveat: a few dozen failures can tune 4 weights, not 16 metrics.
Clustering	Peer cohorts	Learns peer groups from behaviour instead of hand-set rules
Anomaly detection	Patterns outside the 8 diagnoses	Catches a dealer failing in a way nobody listed
Sequence models	Retail and stock, one quarter ahead	Waits until several years of history are in

Never ML: the diagnosis layer stays rules-based, permanently.

Version	What runs	What it needs
1	Percentiles, rules, hard floors, diagnosis table	Tier 1 and 2 data. Available now.
2	NLP, learned weights, clustered cohorts	Lost-dealer record (question 2), text corpus
3	Forecasting, Tier 3 data for flagged dealers	Several years of history, funded collection

Calibration: what we ask the client

The mechanism is ours. The numbers are the client's. Until answered, every screen shows placeholders, labelled as such. Questions 3, 4 and 5 are document requests, which makes those thresholds contractual.

#	Question	What it sets
1	How many dealers can one manager act on in a month?	List length, tier cutoff
2	Which dealers did you lose or nearly lose in 3 years, and what went wrong first?	Weights, and a test of the model on real failures
3	Please share the dealer agreement: credit period and funding terms	Payment delay and ageing floors
4	Please share the stocking norm per dealer, by segment	Inventory floor
5	Please share sanctioned credit and floor plan limits, and the floor plan rate	Credit stress floor, yearly carrying cost
6	How long does a corrective action take to show in the numbers?	Trend window, review cycle
7	How many bad months do you wait before escalating today?	Wobble versus pattern
8	What is contribution per car by segment, and how many buyers of a closed dealer stay with the brand?	Value at stake
9	Which dealer attributes do you treat as comparable?	Peer cohorts
10	Which of these fields do you capture, and how reliable is each?	Which metrics survive version 1

Scope and limits

Prototype scope

- One brand, passenger vehicles, one state, last 18 to 24 months
- District approximated by RTO, since VAHAN reports by RTO, so market opportunity is a directional benchmark, not true dealer catchment
- State coverage shown on every output
- Demand and market run on live VAHAN and FADA data. Customer health runs on real Google reviews for real outlets.
- Finance and inventory use fields that already exist in SAP and DMS, named by system and column, filled with a labelled representative extract

Key limitation: the engine shows association, not cause. It decides where the manager goes. The manager confirms why on the ground.

Also: peer ranking misses a region that falls together. Hard floors and a region-level check cover this.

Sources

- FADA August 2026 retail data
- FADA: inventory 38 to 40 days, 21-day norm
- Autocar Pro: Maruti August 2026 retail
- ICICIdirect: wholesale vs retail, stock by maker
- FADA August 2024 release: Rs 77,800 cr, 7.8 lakh cars
- Deccan Herald: price cuts with GST change
- Motilal Oswal: Maruti FY24 annual report update
- The Machine Maker: Maruti FY26 results
- Maruti: FY26 sales
- Auto Punditz: August 2026 RTO coverage
- Flash Finance: Vahan 4 dashboard coverage
- Dealer Marketing: NADA 45 to 60 days